



STRUCTURAL MECHANICS AND DYNAMICS 3 MECE09036 STRUCTURAL LAB EXPERIMENTS STUDENT REPORT

School of Engineering, The University of
Edinburgh

LAB REPORT

Student name: Annabelle Johnstone
Student surname: Johnstone
Student number: S2502637

Section 1 Structures Lab Exp 1 – Unsymmetric Bending of Beams

1.1 Introduction

10/10

Unsymmetric bending occurs when beams with non-symmetric cross-sections are loaded, causing bending about both principal axes, resulting in combined vertical and horizontal deflection. This behaviour is common in structural members such as angle sections in truss and frame systems. This experiment investigates the unsymmetric bending behaviour of a mild-steel L-section cantilever beam, subject to a vertical end load while incrementally rotating the beam through 180°. Vertical and horizontal tip deflections were recorded at each orientation. Principal bending axes were determined experimentally from the deflection response and compare with theoretical predictions based on second moment of area calculations.

1.2 Apparatus and Methodology

Apparatus

A mild-steel L-section cantilever beam ($E = 210 \text{ GPa}$) was mounted in a rotational fixture fitted with a protractor ($\pm 1^\circ$) to set the beam orientation. Tip deflections in both directions were recorded using digital dial gauges ($\pm 0.01 \text{ mm}$). Cross-sectional dimensions were measured at several points with a micrometre ($\pm 0.01 \text{ mm}$) to capture geometric tolerances. A load of 5.98 N was applied at the free end using calibrated masses and a centroid-loading ring

Methodology

The beam was initially aligned at a 0° , with its legs oriented vertically and horizontally. The dial gauges were zeroed, and a constant load of 5.98 N was applied. Vertical (δ_v) and horizontal (δ_h) deflections were recorded. The beam was then rotated in 10° increments through 180° . At each increment the gauges were re-zeroed, the load applied, and deflections recorded.

Table 1.1: measured beam geometry

Parameters	Mean Value (mm)
Thickness	3.475
Width	25.07
Length	500

Note: Multiple measurements were taken for each parameter to account for manufacturing variations.

1.3 Analysis (theoretical and experimental calculations)

8/10

For a rotated unsymmetric section. Bending moments resolved along principal axes are

$$M_u = PL \sin(\theta - \alpha), \quad M_v = PL \cos(\theta - \alpha)$$

The resulting tip deflections along the principal axes are given by:

$$\delta_u = \frac{M_u L^2}{2EI_v}, \quad \delta_v = \frac{M_v L^2}{2EI_u}$$

These deflections are transformed into measurable geometric coordinate system (y, z) using:

$$\delta_y = \delta_u \sin \alpha + \delta_v \cos \alpha, \quad \delta_z = \delta_u \cos \alpha - \delta_v \sin \alpha$$

The principal axis orientation is calculated from the second moments of area about the geometric axes using:

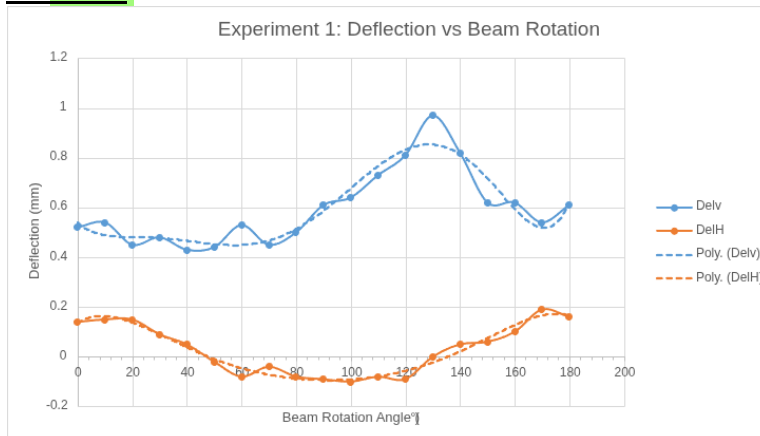
$$\tan 2\alpha = -\left(\frac{2I_{yz}}{I_{yy} - I_{zz}}\right)$$

The analysis assumes linear elastic behaviour and small deflections, allowing superposition and principal axis transformation. This theoretical model will be used to calculate the expected deflections. These values will be

could give a more detailed derivation of the formulas starting from the integral form of the second moment of area ...

compared with the experimental measurements. The detailed calculations are in Appendix A

1.4 Results 8/10



- could mark the theoretical principle axis on the graph.
- what about error bars?

Figure 1.1 experimental deflection vs beam rotation angle

Figure 1 shows the relationship between beam rotation and measured deflection. Both deflection components vary sinusoidally with angle (θ). The horizontal deflection crosses zero at approximately $\theta = 47.1^\circ$ and $\theta = 131.4^\circ$, indicating orientations where a principal axis aligns with the vertical load. The corresponding experimental principal axis angle is therefore:

$$|\alpha_{exp}| = |47.1^\circ|$$

Method	Principal Axis Angle ($^\circ$)	Error (%)
Theoretical	44.6	-
Experimental	47.1	5.4

what about the second principle axis?

1.5 Discussion and Conclusion

Discussion

The 5.4% difference between theoretical and experimental principal axis angle indicates strong agreement with beam-bending theory.

Sources of errors were:

- **Clamp compliance:** Real support not perfectly fixed which leads to increased deflection.
- **Dimensional variation:** Small errors amplified because deflection $\propto L^3$.
- **Protractor accuracy ($\pm 1^\circ$):** Affects angle positioning and zero-crossing detection.
- **Load and dial gauge precision:** Minor offsets influence measurement.
- **10° increments:** Principal axis falls between points, requiring interpolation.

Despite these limitations, results followed expected sinusoidal trends and confirmed the method. The sinusoidal trend in deflection confirms that bending occurred primarily about the rotating principal axes rather than the geometric axes.

What about solutions for improving the sources of errors?

Conclusion

The experiment successfully demonstrated unsymmetric bending behaviour in an L-section cantilever. The principal axes were identified with reasonable accuracy by analysing deflection variation with rotation, validating the theoretical prediction method. Deflection discrepancies highlight sensitivity to boundary conditions and measurement precision; the idealised theory underestimates deflection due to clamp flexibility and material/geometry variation. Overall, the procedure provides a reliable approach for characterising asymmetric beam behaviour.

Appendix A:

8/10

A1. Raw data

Table A.1 Raw data

Angle	Δv	Δh	Δv_{offset}	Δh_{offset}	$\Delta v_{\text{corrected}}$	$\Delta h_{\text{corrected}}$
0	0.52	0.14	0	0	0.52	0.14
10	0.54	0.15	0.06	0.06	0.6	0.21
20	0.45	0.15	0	0.01	0.45	0.16
30	0.48	0.09	0	-0.01	0.48	0.08
40	0.43	0.05	0	0	0.43	0.05
50	0.44	-0.02	0.03	0	0.47	-0.02
60	0.53	-0.08	-0.05	-0.01	0.48	-0.09
70	0.45	-0.04	0.02	0.01	0.47	-0.03
80	0.5	-0.08	0	0	0.5	-0.08
90	0.61	-0.09	0.05	0.02	0.66	-0.07
100	0.64	-0.1	0.01	-0.03	0.65	-0.13
110	0.73	-0.08	0.02	-0.02	0.75	-0.1
120	0.81	-0.09	0.07	0	0.88	-0.09
130	0.97	0	0.05	-0.04	1.02	-0.04
140	0.82	0.05	0.08	0.03	0.9	0.08
150	0.62	0.06	0.05	0	0.67	0.06
160	0.62	0.1	0.03	-0.07	0.65	0.03
170	0.54	0.19	-0.02	-0.03	0.52	0.16
180	0.61	0.16	-0.05	0	0.56	0.16

Table A.2 beam geometry

Dimensions of the beam			
Thickness 1	3.41	3.44	3.45
thickness 2	3.54	3.49	3.54
Width 1	25.44	25.6	25.49
Width 2	24.71	24.73	24.76
Length	500		

what about example calculation of the centroids?

A.2 Section Properties of L-Beam

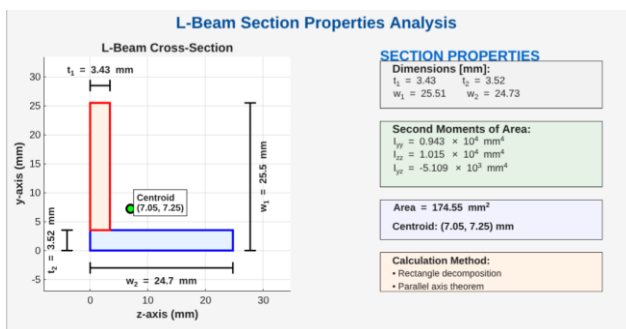


Figure 1.2 section properties

A.3 Principal Axis Orientation

Theoretical principal axis angle is calculated from:

$$\tan 2\alpha = -\left(\frac{2I_{yz}}{I_{yy} - I_{zz}}\right) = \tan 2\alpha = -\left(\frac{2 \times (-4922)}{11520 - 11380}\right) = -\left(\frac{-9844}{140}\right) \approx 70.31 \Rightarrow \alpha_{\text{theo}} \approx 44.6^\circ$$

$$\text{A4. Error } \% \text{Error} = \left| \frac{\alpha_{\text{theo}} - \alpha_{\text{exp}}}{\alpha_{\text{exp}}} \right| \times 100 = \left| \frac{44.6 - 47.1}{47.1} \right| \times 100 \approx 5.4\%$$

A.5 Principal Moments of Inertia

$$I_u = \frac{1}{2} \left[I_{yy} + I_{zz} - \sqrt{(I_{yy} - I_{zz})^2 + 4I_{yz}^2} \right] \approx 1.637 \times 10^{-8} \text{ m}^4$$

$$I_v = \frac{1}{2} \left[I_{yy} + I_{zz} + \sqrt{(I_{yy} - I_{zz})^2 + 4I_{yz}^2} \right] \approx 6.530 \times 10^{-9} \text{ m}^4$$

Note: mm^4 converted to m^4 ($1 \text{ mm}^4 = 10^{-12} \text{ m}^4$)

A.6 Sample Tip Deflection Calculation ($\theta = 90^\circ$)

Applied load: $P = 5.98 \text{ N}$ Beam length: $L = 500 \text{ mm}$

$$M_u = PL \sin(\theta - \alpha) = 5.98 \times 500 \times \sin(90 - (44.6)) = 2192 \text{ N mm}$$

$$M_v = PL \cos(\theta - \alpha) = 5.98 \times 500 \times \cos(90 - (44.6)) = -2099 \text{ N mm}$$

$$\delta_u = \frac{M_u L^2}{2EI_v} = 0.077 \text{ mm}, \quad \delta_v = \frac{M_v L^2}{2EI_u} = -0.192 \text{ mm}$$

Transformed to geometric axes:

$$\delta_y = \delta_u \sin \alpha + \delta_v \cos \alpha = -0.082 \text{ mm}, \quad \delta_z = \delta_u \cos \alpha - \delta_v \sin \alpha = 0.19 \text{ mm}$$

All other angles follow the same calculation method. Note: $\alpha = 44.6^\circ = 0.778 \text{ rad}$, $\vartheta = 90^\circ = 1.571 \text{ rad}$

A.7 Table A.3 Summary of Theoretical and Experimental Deflections

Beam Angle θ ($^\circ$)	δ_y Theo (mm)	δ_z Theo (mm)	δ_y Exp (mm)	δ_z Exp (mm)
0	0.00	0.00	0.00	0.00
30	-0.10	0.12	-0.08	0.15
60	-0.15	0.08	-0.12	0.25
90	-0.17	0.11	-0.09	0.61
120	-0.12	0.15	-0.10	0.55
150	-0.08	0.18	-0.06	0.48
180	0.00	0.20	0.00	0.40

A.8 Assumptions

1. The beam material is linearly elastic, isotropic, and homogeneous.
2. Deflections are small and dominated by bending; shear deformation is neglected.
3. The beam cross-section is prismatic (constant along the length).
4. The fixed support is perfectly rigid and provides full restraint.
5. The load is applied perfectly vertically and at the centroid of the free end

Section 2 Structures Lab Exp 2 – Curved Beam**2.1 Introduction** 9/10

Curved beams develop a non-linear stress distribution because of their geometry, producing higher stresses at the inner radius and lower stresses at the outer radius, unlike straight beams which follow linear bending theory. Curved beams are often seen in hooks, frames and rings. This experiment investigated the load-deflection behaviour of a curved beam by recording vertical and horizontal deflection during incremental loading and unloading. Experimental results were compared with theoretical predictions obtained from Castigliano's theorem to evaluate model accuracy and assess structural stiffness and hysteresis behaviour.

2.2 Apparatus and methodology**Apparatus:**

maybe more details for the "during experiment", like: applying the load carefully to avoid external forces, or taking the reading when stationary (not oscillating)

A mild-steel curved beam ($E = 210 \text{ GPa}$) was tested. Vertical and horizontal crown deflections were recorded using digital dial gauges ($\pm 0.01 \text{ mm}$). The beam's dimensions were measured with a micrometre and a steel rule. The specimen was mounted in a rigid frame and clamped at one end, while calibrated masses applied free-end loads ranging from 2 N to 10 N.

Methodology:

The beam was securely fixed in the test frame with dial gauges positioned at the crown. Load was applied in 2 N increments from 0 N to 10 N, with deflections recorded at each step. The load was then removed in the same increments to assess the unloading response and check for hysteresis. The entire procedure was repeated to verify repeatability of measurements.

2.3 Analysis 5/10

The deflection of the curved beam was calculated using results derived from Castigliano's second theorem for semicircular beams under point loads.

$$\delta_v = \frac{\pi PR^3}{2EI}, \delta_h = \frac{2PR^3}{EI}$$

The second moment of area for a rectangular cross section is given by $I = \frac{bt^3}{12}$

Where the Elastic stiffness defined by $k = \frac{P}{\delta}$

equations (bending moment, strain energy integral) and brief derivations, etc ??

$$k_v = \frac{2EI}{\pi R^3}, \quad k_h = \frac{EI}{2R^3}$$

The theoretical predictions are based on linear elastic small-deflection beam theory, assuming material isotropy, negligible shear deformation, and a perfectly fixed support. The experimental stiffness was determined from the gradient of the load-deflection graphs, a method that eliminates errors from initial measurement offsets. Detailed calculations are provided in Appendix B.

2.4 Results

The mean load-deflection values from two test runs are shown in Figure 2.1 and 2.2.

you need the average of loading & unloading deflections as the experimental deflection (and fit a linear func ...) and should've had the theoretical ones in the same graph

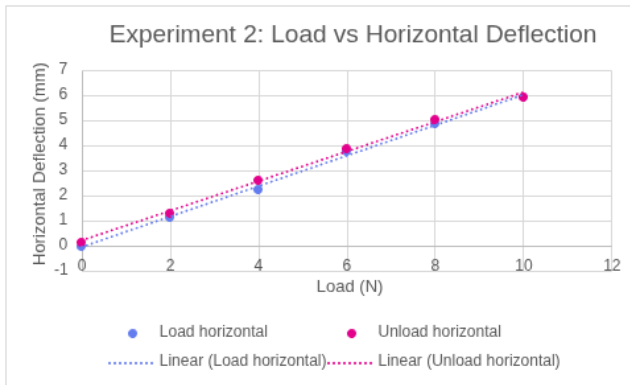


Figure 2.1: load vs vertical deflection

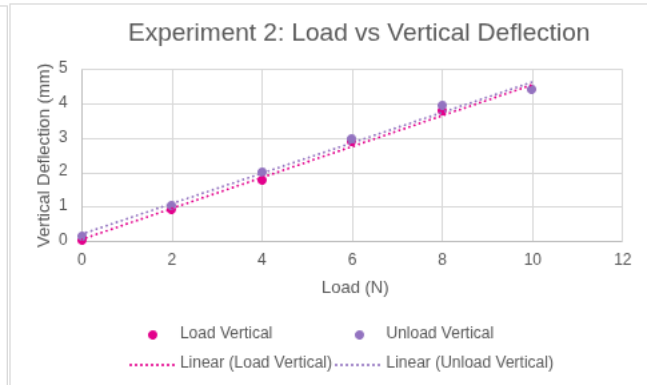


Figure 2.2: load vs horizontal deflection

Horizontal deflections exceeded vertical deflections at all loads, as predicted by curved beam theory. Minimal separation between loading/unloading paths indicating elastic behaviour with negligible hysteresis. Small zero-load offsets indicated minor system compliance and gauge drift.

Table 2.1- stiffness comparison and error

Direction	Theoretical Stiffness (N/mm)	Experimental Stiffness (N/mm)	Error
Vertical	3.29	2.30	29.3%
Horizontal	2.60	1.72	33%

based on loading or unloading or the average?
(not mentioned)

2.5 Discussion and conclusion

Discussion 6/10

The curved beam demonstrated a linear and elastic load-deflection response, as evidenced by the minimal hysteresis, confirming the fundamental material assumption. However, the experimental deflections were consistently smaller than theoretical predictions, resulting in the experimental stiffness being 29-33% lower than theoretically calculated.

These discrepancies can be attributed to:

- **Support compliance:** The theoretical model assumes a perfectly rigid fixed end; real clamps allow slight movement.
- **Geometric sensitivity:** Deflection scales with R^3 ; small radius measurement errors amplify results.
- **Measurement resolution:** Small deflections increase sensitivity to dial-gauge accuracy and human reading error.
- **Frame friction/misalignment:** Minor external effects not represented in theoretical model.

Castigliano's theorem provided correct behavioural trends but underestimated actual deflections due to idealised assumptions. The stiffness-based analysis proved more reliable by eliminating zero-offset errors.

Conclusion

The experiment successfully demonstrated the fundamental load-deflection behaviour of a curved beam. The theoretical model provided a good qualitative prediction of the beam's response. However, the quantitative discrepancy in stiffness highlights the critical influence of real-world boundary conditions and measurement precision. This underscores the importance for engineers to recognise the limitations of analytical models and account for practical constraints when applying them to the design and analysis of real structures.

specific values? quantitative comparison with the theoretical values (percentage difference)??
What about solutions for improving the sources of errors?

Appendix B: 9/10**B.1 Raw Data Sheet**

Two runs were recorded for loading and unloading to check repeatability. Mean values were used for analysis.

Table B.1 Raw Deflection Measurements

Load (N)	Run1 H load	Run1 H unload	Run2 H load	Run2 H unload	Run1 V load	Run1 V unload	Run2 V load	Run2 V unload
0	0.00	0.18	-0.01	0.18	0.00	0.12	0.00	0.12
2	1.10	1.32	1.21	1.31	0.87	1.02	0.96	1.01
4	2.10	2.68	2.40	2.54	1.66	2.03	1.89	1.99
6	3.79	3.97	3.78	3.78	2.81	2.95	2.97	2.98
8	5.15	5.07	4.63	5.05	3.87	3.87	3.75	3.99
10	6.01	6.01	5.83	5.83	4.08	4.08	4.72	4.72

Note: Zero-load readings include gauge seating drift and fixture compliance.

Table B.2 Beam dimensions

Dimension	First Meas. (mm)	Second Meas. (mm)	Third Meas. (mm)	Mean (mm)
Thickness (t)	3.4	3.3	3.4	3.37
Width (b)	25.0	25.2	25.1	25.1
Beam Diameter (D)	297	298	297	297.3
Mean Radius (R)				148.65

Table B.3 Final mean deflection

Load (N)	Final Mean Horizontal (mm)	Final Mean Vertical (mm)
0	0.0875	0.060
2	1.235	0.965
4	2.430	1.893
6	3.830	2.928
8	4.975	3.870
10	5.920	4.400

B.2 Material and section Properties

The **second moment of area** for a rectangular cross-section is:

$$I = \frac{bt^3}{12}, \quad b = 25.1 \text{ mm (width)}, \quad t = 3.37 \text{ mm (thickness)}$$

$$I = \frac{25.1 \times (3.37)^3}{12} = 80.05 \text{ mm}^4$$

Other constants: $R = 148.65 \text{ mm}$, $E = 210 \times 10^3 \text{ N/mm}^2$

Note: Properties calculated assuming thin rectangular section, bending about centroidal axis.

B3. Stiffness calculation (gradient method)

Using extreme points:

$$k = \frac{P}{\delta} \quad k_{exp,v} = \frac{10}{4.40 - 0.06} = 2.30 \text{ N/mm} \quad k_{exp,h} = \frac{10}{5.92 - 0.09} = 1.72 \text{ N/mm}$$

Theoretical stiffness

$$k_v = \frac{2EI}{\pi R^3} = 3.26 \text{ N/mm}, \quad k_h = \frac{EI}{\pi R^3} = 2.30 \text{ N/mm}$$

B3. Stiffness calculation error

$$\%Error = \left| \frac{k_{theo} - k_{exp}}{k_{theo}} \right| \times 100$$

Table B.3 – Stiffness Error

Direction	k_{theo} (N/mm)	k_{exp} (N/mm)	% Error
Vertical	3.26	2.30	29.3%
Horizontal	2.56	1.72	33%

example calculation of theoretical deflection??

B.5 Assumptions

1. The beams material is linearly elastic, isotropic, and homogeneous.
2. Strain energy is dominated by bending; shear deformation is neglected.
3. The beam is a perfect circular arc with constant cross-section.
4. The supports are perfectly rigid; fixed end fully restrained, free end free to deflect.
5. Curved beam theory applies, small deformation theory.
6. Dial gauges zeroed before each load cycle; temperature effects negligible.